

T03 THERMAL ZONING | O (MAX : 2 PT)

Intent : Enhance thermal control of building occupants through the provision of thermal zones.

Summary : This WELL feature requires projects to increase thermal control of the space, by allowing control of either the conditions of a thermal zone or movement between thermal zones.

Issue : People in buildings who are satisfied with their thermal environment have been shown to be more productive in the workplace.^{1,2} For instance, employees perform on average 15% poorer in hot conditions and on average 14% poorer in cold conditions.³ An average 2% decrease in work performance per 1 °C of air temperature rise has been found, when the air temperature is above 25 °C.⁴ On the other hand, even a 1 °C decrease in air temperature within the range of 20 °C-25 °C is linked to increased student performance in mathematics.⁵ Furthermore, thermal comfort preferences are highly individual and can be affected by metabolism, body type, clothing and other personal factors.⁶⁻⁸ These personalized factors make it nearly impossible to find a temperature that will satisfy all individuals in the same space at the same time.⁹

Solutions : Indoor air temperature can be influenced by a number of factors, including the building physics and orientation, building location and structure, occupant density, ventilation strategy and mode of operation. In the majority of buildings, the HVAC system is responsible for controlling indoor air temperature and humidity. Where temperature zoning is an option, individually accessible thermostats that enable people to set their own thermal conditions independently of other zones should be used.

PART 1 PROVIDE THERMOSTAT CONTROL (MAX : 2 PT)

For All Spaces:

The following requirements are met for at least 90% of regularly occupied spaces:

- a. Control over temperature in the space is available through either:
 1. Thermostats present within the thermal zone.
 2. A digital interface accessible to occupants on a computer or phone.
- b. The maximum size of each thermal zone is 60 m² or 10 occupants, whichever is larger. Individual regularly occupied rooms are considered separate zones.

c.

Projects earn points based on the number of thermal zones:

Tier	Number of Zones		Number of Zones	Points
1	One per 60 m ²	OR	One per 10 occupants	1
2	One per 30 m ²	OR	One per 5 occupants	2

- d. Temperature sensors are positioned at least 1 m away from exterior walls, windows and doors, direct sunlight, air supply diffusers, mechanical fans, heaters or any other significant sources of heat or cold.

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